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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1 **Date:** September 19 2026

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just “programming.” As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, developing, testing, and maintaining software systems.	Software Developer, Software Engineer
Systems & Cloud Infrastructure	Managing computer systems, networks, servers, and monitoring performance traces to maintain real-time visibility into the health of the cloud estate.	Systems Administrator, Cloud Engineer
Data Science & AI Engineering	Collecting and analyzing data and developing AI and machine learning solutions.	Data Scientist, AI Engineer
Cybersecurity & Threat Defense	Protecting computers, networks, systems, and data from cyber threats and attacks like viruses that are designed to harm or gain unauthorized access to a system.	Cybersecurity Analyst, Security Engineer
Information Systems & Enterprise IT	Managing technology systems that support business operations and organizational needs.	IT Specialist, Systems Analyst

My Chosen Pathway (Part B)

I am most interested in:

Why I chose it:

I am most interested in Cybersecurity & Threat Defense because, with the increasing use of technology today, there are many threats and cyber attacks, such as malware, viruses, and other harmful activities that can put people's privacy and information at risk. I am interested in this pathway because I want to learn how to protect personal data, computer systems, and networks from these threats. I also want to develop my skills in cybersecurity and understand how cyberattacks happen so I can help prevent them and make technology safer for everyone.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Cybersecurity Analyst, IT Security Technician	Monitor systems, identify threats, protect data, and learn basic cybersecurity tools.
Mid-Level	Security Engineer, Incident Response Analyst	Investigate security incidents, improve security systems, and respond to cyber attacks.
Senior-Level	Senior Cybersecurity Analyst, Senior Security Engineer	Lead security investigations, manage security projects, and develop strategies to protect systems and networks.
Expert / Leadership	Security Architect, Chief Information Security Officer (CISO)	Design security strategies, lead cybersecurity teams, manage major security risks, and make important security decisions.

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Basic cybersecurity concepts	N	I will continue studying cybersecurity topics and practice defenses Cybersecurity attacks.
Networking fundamentals	N	I will study networking through school lessons, tutorials, and hands-on exercises.
Threat and malware detection	N	I will learn how different cyber threats work and practice identifying them through safe activities.
Programming / scripting	Y	I will continue practicing Python and other programming languages in my classes.
Problem-solving and critical thinking	Y	I will improve these skills by solving programming and cybersecurity exercises.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Cybersecurity Analyst (Entry-Level / Graduate)
Company	Datamatics
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	LinkedIn
Required Skills Listed (3–5)	1. Basic cybersecurity knowledge 2. Knowledge of common cyber threats and vulnerabilities 3. Basic network and web security knowledge 4. Knowledge of cloud security concepts 5. Problem-solving and analytical skills

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I chose Cybersecurity & Threat Defense because I am interested in protecting people, data, and computer systems from cyber threats. In the modern world, where all data is digitalized, such threats are one of the security issues that need to be addressed. Therefore, I want to learn how to mitigate and avoid them.

2. Looking at your roadmap, which career stage excites you the most, and why?

The Expert/Leadership stage is the most exciting for me due to the increased amount of knowledge and experience in the field of cybersecurity that one gets at this level. I also want to lead and manage a team of experts and develop new methods of protecting data and systems from attacks.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I already have some basic skills and knowledge of cybersecurity awareness which I can apply while studying cybersecurity. The most challenging skills for me to acquire are ethical hacking and network security due to the high level of technical expertise needed. I intend to learn these skills through adequate practice and study.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

This roadmap assures me that I want to learn more about cybersecurity throughout my course and degree. It inspires me to develop my programming, networking, and problem-solving skills. I also desire to gain adequate experience and knowledge to be a Cybersecurity Analyst in the future.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date